10 States of Matter

SOLUTION (QR CODE)

3. (B) Decreases by 20%

$$PV = \frac{5P}{4 \times V'}, \quad V' = \frac{4V}{5}$$

$$\% \text{ decrease in volume} = \frac{V - \frac{4V}{5}}{V} \times 100\% = 20\%$$

4. $T_1 = 27 + 273 = 300$ K

$T_2 = 57 + 273 = 300$ K

Now, $\frac{V_1}{T_1} = \frac{V_2}{T_2}$, $\frac{1}{300} = \frac{V}{330}$

$V = \frac{330}{300}$ L = 1.1 L

6. Volume of balloon $= \frac{4}{3}\pi r^3 = 4 \cdot 851$ lt

Calculate final volume (V_2) as:

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \Rightarrow V_2 = 51.32$ lt

Volume of gas available for filling of balloons=51.32-2.82=48.50 L

No. of balloons= 48.50/4.851 = 10

9. First calculate partial pressure of dry gas:

$P_{dry\ gas} = 4 - 0.4 = 3.6$ atm

Apply Boyle's law:

$P_1V_1 = P_2V_2$

$3.6 \text{ atm} \times V_1 = P_{dry\ gas\ (new)} \times 2V_1$

$P_{dry\ gas\ (new)} = \frac{3.6 \text{ atm}}{2} = 1.8 \text{ atm}$

Now, calculate pressure of moist gas:

$P_{moist\ gas\ (new)} = P_{dry\ gas\ (new)} + P_{water\ vapor}$

$= 1.8 \text{ atm} + 0.4 \text{ atm} = 2.2 \text{ atm} = 22 \times 10^{-1}$ atm

Dimag ghum toh nhi gaya!

17. $P_{H_2} = \frac{2}{2+3+4} \times 800$ torr

$P_{NH_3} = \frac{3}{2+3+4} \times 800$ torr

$P_{CO_2} = \frac{4}{2+3+4} \times 800$ torr

On solving,

$P_{H_2} = 178$ torr, $P_{NH_3} = 267$ torr,

$P_{CO_2} = 356$ torr

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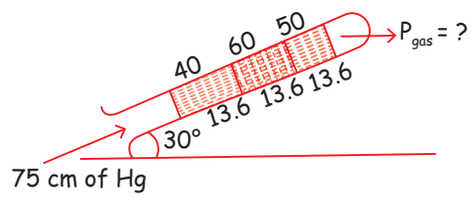
20. (B) 65 cm of Hg $\Rightarrow \frac{65}{76}$ atm

$\Rightarrow \frac{65}{76} \times 1.013 \times 10^5$ pascal

$\Rightarrow \frac{65}{76} \times 1.013 \times 10^5 \text{ N/m}^2$

Agar convert nahi karna aata toh ghumte reh jaoge

24. (B)



$75 = 150 \sin 30° + P_{gas}$

$75 - 150 \sin 30° = P_{gas}$

$P_{gas} = 0$

So, barometer is not faulty.

29. $\frac{r_{N_2}}{r_{Unknown}} = \frac{1/38}{1/56.2} = \frac{P_1}{P_2}\sqrt{\frac{M_2}{M_1}} = \frac{0.8}{1.6}\sqrt{\frac{M}{28}}$

$\frac{56.2}{38} = \frac{0.8}{1.6}\sqrt{\frac{M}{28}}$

M = 245

Let the formula of compound be XeF_n.

131 + 19n = 245, n = 6

Hence, the molecular formula of the compound is XeF_6.

acha sawaal hai bacho, dhyan lagao

36. (i) $P = \frac{1}{3}\frac{mN\overline{u}^2}{V} = \frac{1}{3} \times \frac{10^{-25} \times 10^{23} \times 10^6}{10^{-3}}$

Vol in SI unit in m^3

$\overline{u}^2 = \left(10^3\right)^2$

$P = \frac{1}{3} \times 10^7$ Pascals

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(ii) $\frac{1}{2}m\overline{u^2} \times$ no of particles = Total K.E.

$\frac{1}{2} \times 10^{25} \times 10^6 \times 10^{23}$ = 5000 J

= 1195 cal

(iii) 1195 = $\frac{3}{2} \times R \times T \times$ Moles

$T = \frac{2 \times 1195 \times 6.022 \times 10^{23}}{3 \times 1.98 \times 10^{23}}$

= 2415 K

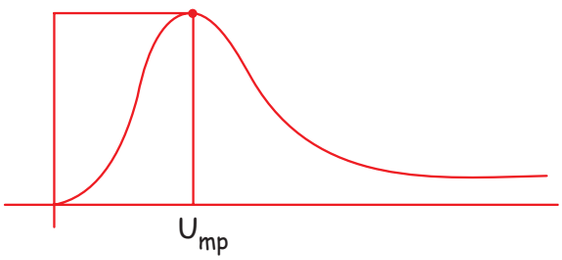
38. Root mean sq speed of X = most probable of Y

$\sqrt{\frac{3 \times R \times 400}{40 \times 10^{-3}}} = \sqrt{\frac{2 \times R \times 60}{M \times 10^{-3}}}$

$\frac{3 \times R \times 400}{40 \times 10^{-3}} = \frac{2 \times R \times 60}{M \times 10^{-3}}$

$M = \frac{120}{30} = 4$

41. (A) Velocity dist. function is max means



$U_{mp} = 400 = \sqrt{\frac{2 \times R \times T}{32 \times 10^{-3}}}$

T = 34.9° C

42. $4\pi\left(\frac{M}{2\pi RT}\right)^{3/2} C_1^2 \cdot e^{\frac{-MC_1^2}{2RT}} = 4\pi\left(\frac{M}{2\pi RT}\right)^{3/2} C_2^2 \cdot e^{-\frac{MC_2^2}{2RT}}$

$\frac{C_1^2}{C_2^2} = e^{-\frac{M}{2RT}\left[C_2^2 - C_1^2\right]}$

$\ln\frac{C_1^2}{C_2^2} = \frac{M}{2RT}\left[C_1^2 - C_2^2\right]$

On solving, T = 64.5°C.

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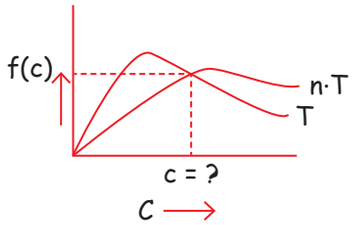
43. (A)



$4\pi\left(\frac{M}{2\pi RT}\right)^{3/2} \times c^2 \times e^{-\frac{MC^2}{2RT}}$

$= 4\pi\left(\frac{M}{2\pi RT.n}\right)^{3/2} \times c^2 \times e^{-\frac{MC^2}{2RT.n}}$

or, $n^{3/2} = e^{\frac{MC^2}{2RT}\left(1-\frac{1}{n}\right)}$ or, $\frac{3}{2}\ln n = \frac{MC^2}{2RT} \times \frac{n-1}{n}$

$\therefore\ C = \sqrt{\frac{3nRT\ln n}{M(n-1)}}$

52. Van der waal equation is:

$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$

Substitute the values in the above expression:

$\left(11 + \frac{a \times 4}{16}\right)(4 - 2 \times 0.05) = 2 \times 0.082 \times 300$

$\left(\frac{176 + 4a}{16}\right) \times 3.9 = 49.2$

(176 + 4a) 3.9 = 49.2 x 16

On solving, a = 6.46 atm Lt^2 mol^{-2}

Kch nhi karna bs vander waal equation use hogi

55. The formula to calculate compressibility factor (Z) is:

Z = PV/RT

According to van der waal equation:

$$\left(P+\frac{a}{V^2}\right)(V-b)=RT$$

According to given information, b is negligible that is:

$$\left[100+\frac{a}{(0.112)^2}\right]=0.0821\times 273$$

On solving, a = 1.253 L^2 mol^{-2} atm

60. $b=4\times\frac{4}{3}\pi r^3 N_A=\cancel{4}\times\frac{\cancel{4}}{3}\cancel{\pi}\frac{100}{\cancel{16\pi N_A}}\cdot\cancel{N_A}=\frac{100}{3}$

$$V_c=3b=\frac{100}{3}\times 3=100cc$$

$$d_c=\frac{M}{V_c}=\frac{44}{100}=0.44g/cc.$$

64. $NH_3 + H_2 \xrightarrow{\frac{3}{2}} N_2 + H_2 + H_2$

mm O_2 (40 ml)

$H_2O + O_2$ left 6 ml

NH_3 + H_2 → Present initially

x ml (50 – x) ml

$$\%\text{ of } NH_3=\frac{36}{50}\times 100 = 72\%$$

$$NH_3\rightarrow N_2+\frac{3}{2}H_2$$

x ml $\frac{3}{2}H_2$ (Formed after decomposition)

$$1H_2+\frac{1}{2}O_2\rightarrow H_2O$$

To consume 34 ml of O_2 the requirement of H_2 is 68 mL.

Total H_2 in the tube = $\left(50-x+\frac{3x}{2}\right)$ ml

x = 36 ml

isko chhod mat dena , samjhe

66. (D) The general combustion reaction in terms of x and y is:

$$C_xH_y+\left(x+\frac{y}{4}\right)O_2\rightarrow xCO_2+\frac{y}{2}H_2O$$

The volume of oxygen required to react with the hydrocarbon is

$$\left(x+\frac{y}{4}\right)\text{ volume of } C_xH_y=\left(x+\frac{y}{4}\right)\times 12$$

On passing KOH, CO_2 gets absorbed. The volume left is only of O_2 which is 8 mL.

$$\left\{50-12\left(x+\frac{y}{4}\right)\right\}+12x=32$$

On solving we get that the

$$y=\frac{50-32}{3}=6$$

Now we will write equation with only oxygen left

$$50-12\left(x+\frac{y}{4}\right)=8$$

On solving we get, 2x = 32 – 8 = 2

Formula of hydrocarbon= C_2H_6

Yeh thoda tedha sawaal hai lekin samjh ajaega

67. (B) The balanced equation for the combustion reaction is:

$C_4H_x + (4 + x/4)O_2 \rightarrow 4CO_2 + x/2H_2O$

Since 10 mL of C_4H_x requires 55 mL of O_2 for complete combustion, we can set up the following equation:

(4 + x/4) = 55/10, (4 + x/4) = 5.5

Simplifying the equation, we get:

x/4 = 1.5, x = 6